

The Impact of Agentic AI on Real-Time Credit Risk Assessment and Dynamic Limit Allocation

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Abstract—Credit risk assessment is important for banks and financial institutions because digital banking, online lending, and card-based payments create borrower activity at high speed. Traditional credit scoring systems mostly depend on past records, fixed credit reviews, and delayed policy updates. These systems can detect risk, but they often react too late. This creates a gap between risk prediction and credit limit adjustment. This paper proposes an Agentic AI framework for real-time credit risk assessment and dynamic credit limit allocation. The proposed framework uses a monitoring agent, preprocessing layer, SMOTE balancing, feature engineering, XGBoost reasoning, explainability support, and a policy-based action layer. The UCI Default of Credit Card Clients dataset was used for testing. The model achieved 79% accuracy and an ROC-AUC score of 0.88. The confusion matrix, feature importance graph, and dynamic limit simulation show that recent repayment behavior is the most useful signal for credit risk. The framework shows how prediction can be connected with action. Future work may extend this system using live transaction streams, federated learning, graph neural networks, and stronger verification methods.

Index Terms—Agentic AI, Credit Risk Assessment, Dynamic Credit Limit Allocation, XGBoost, SMOTE, Explainable AI.

I. INTRODUCTION

Credit risk assessment is one of the most important functions in banking and financial technology. A bank must decide whether a borrower is likely to repay money on time. If this decision is wrong, the bank may suffer financial loss. If the decision is too strict, good borrowers may lose access to useful credit. This balance has become more difficult in modern digital finance because borrower behavior changes very quickly. Online lending platforms, mobile wallets, credit cards, digital payments, and instant loan services allow customers to make financial decisions at a much faster pace than before. Because of this change, financial institutions need systems that can assess borrower risk in near real time instead of waiting for slow monthly or quarterly reviews [1].

Traditional credit scoring systems were mainly designed for slower banking environments. In these systems, a borrower is evaluated using past repayment history, income records, credit reports, and demographic variables. These records are useful, but they usually describe what happened in the past. They do not always capture sudden changes in borrower behavior. For example, a customer may have a good credit record, but their recent repayment behavior may become weak because of overspending, salary delay, job loss, emergency expenses, or market conditions. If the bank waits until the next review

cycle, the borrower may continue using a high credit limit even after the risk has increased. This increases the chance of default and raises the financial exposure of the institution.

Dynamic credit limit management tries to solve this problem by changing the credit limit when borrower risk changes. This means the system does not only predict whether a borrower is risky. It also decides how much credit should be made available after the risk score is calculated. This is important because risk management is not useful if it stops at prediction only. A bank needs to convert the prediction into an operational action. If risk is low, the limit may be maintained or increased. If risk is high, the limit may be reduced or sent for manual review. Research on dynamic credit limit management shows that adaptive decisions can help institutions manage borrower exposure more effectively [2].

Agentic AI is useful for this problem because it separates a complex financial process into smaller intelligent components. In a normal machine learning pipeline, data is processed, a model gives a prediction, and the output is reported. In an agentic design, different agents perform different roles. A monitoring agent can observe borrower behavior. A reasoning agent can calculate the probability of default. An explainability agent can show why the model made a decision. An action agent can adjust the credit limit. A governance layer can check whether the action is safe and allowed. This structure is useful for finance because financial decisions require both intelligence and control. Recent studies on agentic AI in finance explain that agent-based systems can improve automation, but they must be designed with strong safety, explainability, and compliance rules [3].

Real-time credit scoring is also becoming important because lending platforms are now expected to make quick decisions. A borrower may apply for a loan online and expect a decision within minutes. A credit card system may need to detect risk while transactions are happening. A bank may need to respond to repayment behavior before the borrower defaults. Real-time scoring systems support faster decisions, but many of them still focus only on producing a score. They do not always show how that score should be used for limit allocation, policy checking, and audit reporting [4]. This creates an important gap between machine learning prediction and actual banking decision execution.

Automated underwriting and retrieval-based financial sys-

tems have also been studied in recent finance research. These systems show that AI can help collect information, read documents, check compliance, and support loan decisions [5]. However, underwriting support is not the same as continuous credit limit adjustment. A dynamic credit limit system must keep checking borrower behavior even after the original loan or card approval. It must also react when borrower risk changes. This requires an architecture that works continuously instead of only at the time of application.

Agentic AI architectures for financial scenarios suggest that financial AI systems should be evaluated not only by accuracy, but also by their ability to coordinate tasks and support reliable decisions [6]. This is very important in credit risk assessment. A model may have strong accuracy, but it may still be unsafe if it gives unexplained decisions or if it directly changes a customer limit without policy control. A strong framework must include monitoring, prediction, explanation, action validation, and reporting. This paper follows this idea by treating credit risk as a complete decision process rather than a single classification task.

Explainable AI is also central to this research. Credit decisions affect real people. If a bank lowers a customer's limit, rejects a loan, or marks a borrower as risky, the institution must be able to explain the decision. Methods such as LIME and SHAP are useful because they show which features had the most effect on the model output [7]. In this study, explainability is included as a separate agent because transparency is not optional in finance. It is part of responsible and practical system design.

This study proposes an Agentic AI framework for real-time credit risk assessment and dynamic credit limit allocation. The framework uses the UCI Default of Credit Card Clients dataset, applies SMOTE to handle class imbalance, uses feature engineering to identify important predictors, trains an XGBoost model as the reasoning agent, and simulates dynamic credit limit changes using risk-based logic. The purpose is to show how borrower monitoring, risk prediction, explanation, and credit limit action can work together in one structured framework.

A. Main Contributions and Novelty

The main contributions of this study are as follows:

- Propose an Agentic AI framework that connects credit risk prediction with dynamic credit limit allocation.
- Design a layered architecture with monitoring, preprocessing, reasoning, explainability, governance, and action components.
- Develop an XGBoost-based reasoning model for borrower risk classification using structured credit card client data.
- Analyze the results using accuracy, precision, recall, F1-score, confusion matrix, ROC-AUC curve, and feature importance.
- Demonstrate a simulated credit limit adjustment process that reduces borrower exposure when predicted risk increases.

The remainder of this paper is organized as follows. Section II explains the research gap. Section III reviews related literature. Section IV presents the problem statement. Section V explains the proposed methodology. Section VI presents the system architecture and workflow. Section VII discusses implementation details. Section VIII presents results and discussion. Section IX explains complexity, security, and privacy analysis. Section X presents limitations. Section XI presents future work, and Section XII concludes the paper.

II. RESEARCH GAP

Most existing credit risk studies focus on prediction accuracy. They train a model and report metrics such as accuracy, precision, recall, or ROC-AUC. This is useful, but it is not enough for a real financial institution. In a bank, the main question is not only whether a borrower is risky. The main question is what the bank should do after the borrower is marked risky. A useful system must connect prediction with action. Current research often separates these two steps.

Another gap is the delay in traditional credit systems. Many credit scoring systems still depend on batch processing. This means borrower information is collected and reviewed after a fixed period. In fast digital finance, this delay is risky. Borrower behavior can change before the next review takes place. If the system cannot react quickly, the bank may keep giving credit to a borrower whose risk has already increased.

There is also a gap in autonomous execution. Many studies build machine learning models, but they do not include an action layer. The model gives a prediction, but there is no clear process for changing the credit limit, sending the case for manual review, or creating an audit report. This makes the model less useful in real banking operations. The proposed framework addresses this by adding an Action Agent and a Governance Layer.

Explainability is another major gap. High-performing models are often difficult to interpret. In finance, a black-box system is risky because credit decisions must be justified. If a customer asks why their limit was reduced, the bank needs a clear explanation. If an auditor checks the system, the decision must be traceable. Many existing systems add explainability after the model is built, but they do not make it a central part of the system. This study treats explainability as a core agent in the framework.

Governance is also missing in many credit risk models. A model should not be allowed to change a credit limit without policy checking. There must be safety rules, minimum limits, threshold controls, and audit logs. Recent work on finance-grade assurance for agentic AI explains that governance is necessary when AI systems are used in high-stakes financial environments [8]. The proposed framework includes a policy guard so that actions are validated before execution.

Multi-agent risk management research also shows that complex risk environments require coordinated components rather than one isolated model [9]. Credit risk assessment is similar because the system must handle data, prediction, explanation, policy, and action. These functions are connected, but they

are not the same. This research fills the gap by designing a coordinated agent-based framework for credit risk and credit limit management.

Distributed multi-agent system research highlights the importance of risk control when several autonomous components work together [10]. If one part of the system makes a wrong decision, it may affect the final action. This means the system needs safety checks and monitoring. In the proposed framework, the Action Agent is not allowed to directly execute changes without passing through the policy guard.

Large surveys of agentic AI show that agent-based systems can support planning, reasoning, and decision-making across different domains [11]. However, applying agentic AI to credit risk requires domain-specific controls because banking decisions affect money, customers, and regulation. General agentic AI ideas are not enough. They must be adapted to the financial context.

Systematic research on agent-based hybrid AI models shows that hybrid systems can combine machine learning, rules, human review, and automated decision support [12]. This supports the design used in this paper. The proposed system does not rely only on machine learning. It combines model prediction with SMOTE balancing, feature selection, explainability, and governance logic.

The main research gap can therefore be stated clearly. Existing credit risk systems do not fully integrate real-time monitoring, machine learning risk prediction, explainability, governance validation, and dynamic credit limit action in one framework. This study addresses that gap by proposing and testing an Agentic AI framework for credit risk assessment and dynamic limit allocation.

III. LITERATURE REVIEW

A. Machine Learning and Deep Learning for Credit Risk

Credit risk prediction has changed from simple statistical models to advanced machine learning and deep learning systems. Earlier credit systems used methods such as logistic regression, decision rules, and scorecards. These methods were easy to explain, but they were limited when borrower behavior became more complex. Modern data contains many variables, including repayment status, billing amount, payment amount, credit limit, age, education, and marital status. Machine learning models are useful because they can learn patterns from these variables.

The HFTCRNet study used a hierarchical fusion transformer for interbank credit rating and focused on how risk can move across financial networks [13]. This kind of work is useful because it shows that credit risk is not always isolated. A borrower, firm, or institution may be affected by connected financial behavior. However, transformer models are usually complex and may need more computational resources. They may also be difficult to explain to non-technical banking staff.

Neuromorphic computing research for risk assessment focuses on ultra-low latency financial decisions [14]. Low latency is important because some financial decisions must be made quickly. In credit risk, speed matters when borrower risk

changes rapidly. However, neuromorphic methods may require specialized hardware and may not be easy to implement in normal banking systems. This paper uses XGBoost because it is easier to implement and more suitable for structured tabular credit data.

Deep learning credit scoring research shows that neural models can improve prediction in the financial sector [15]. Deep learning can capture complex relationships, but it also creates explainability problems. In credit risk, a highly accurate model is not enough if the institution cannot explain its decisions. This is why this paper uses a simpler and more interpretable tree-based model rather than a very deep neural network.

Audit data analysis using generative AI is also relevant because financial systems need better tools for risk identification and reporting [16]. Generative AI can help summarize audit trails and support compliance work. In the proposed framework, the audit function is represented through decision logs and explainability outputs. This allows every credit limit decision to be recorded and reviewed.

Ensemble models and explainable AI have also been applied to credit risk assessment [17]. Ensemble models are useful because they combine multiple decision patterns. XGBoost is an ensemble method that builds decision trees step by step to reduce prediction error. It is suitable for credit risk because the dataset is structured and contains nonlinear relationships.

Explainable credit scoring models support user empowerment because they help customers and institutions understand why a decision was made [18]. This is important because credit decisions are sensitive. If a borrower is classified as risky, the system should show the main reasons, such as recent repayment delay or high billing amount. The proposed Explainability Agent supports this requirement.

B. Federated Learning, FinTech Stability, and Legal Risk

Federated graph learning has been studied for dynamic modeling of risk correlation networks among financial institutions [19]. This is useful because banks may want to improve risk models without sharing raw customer data. Privacy is a serious issue in finance. Future versions of the proposed framework can use federated learning so that several banks can improve the model without exposing customer records.

FinTech innovation can improve financial access, but it can also create new stability risks [20]. Faster lending and digital credit systems make financial services easier to use, but they also increase the speed at which credit risk can grow. This supports the need for real-time monitoring. If lending becomes faster, risk management must also become faster.

Explainable machine learning studies on bank credit worthiness show that prediction must be combined with interpretation [21]. Banks cannot rely only on accuracy. They need to know why a borrower was classified as safe or risky. This idea is also supported in the proposed framework because the system includes feature importance and explainability mechanisms.

Personal credit risk prediction and legal prevention research shows that credit systems must consider legal and regulatory issues [22]. A credit model that works technically may still fail if it violates customer rights, privacy rules, or fairness expectations. This is why the proposed framework includes a governance layer and policy guard.

Machine learning research on credit risk assessment for medium and micro enterprises also shows that borrower type matters [23]. Small firms and individual borrowers may have different data patterns. A model trained on one type of borrower may not perform equally well on another. This is one reason future work should test the framework on more datasets.

A second personal credit risk prediction study also emphasizes the connection between prediction and legal prevention [24]. This supports the idea that financial AI must not only predict risk, but also prevent unsafe decisions. The policy guard in this framework is designed for that purpose.

Machine and deep learning research for compliant credit scoring shows that compliance must be included in model design [25]. A model is not truly useful if it cannot be used in a regulated environment. This is why this paper does not only present a classifier. It presents a full framework with monitoring, action, explanation, and governance.

C. Advanced Financial Systems and Alternative Data

Blockchain-enabled finance and warehouse financing research shows how financial systems can use technology to improve trust, transparency, and process efficiency [26]. Although this paper does not implement blockchain, the idea of transparent records is relevant. The proposed audit ledger can be extended in future work using tamper-resistant record systems.

GNN robustness research in financial data shows that financial systems must handle data drift [27]. Data drift happens when the data pattern changes over time. This is common in finance because borrower behavior changes during inflation, recession, unemployment, or market shocks. Future versions of this framework can use drift detection to update the model when the borrower population changes.

Systematic research on LLM applications in industry shows that language models are increasingly used for decision support, document analysis, and knowledge workflows [28]. In credit risk, LLMs may support compliance reports or customer explanations. However, LLMs should not directly make high-stakes credit decisions without verification. In this paper, the main decision model remains XGBoost.

Critical analysis of AI in financial risk management highlights that AI systems can improve risk detection, but they also create new risks when they are not controlled [29]. This supports the governance design in the proposed framework. Autonomous systems should not be treated as fully trusted without policy checks.

Multi-agent reinforcement learning for dynamic resource allocation is also relevant because credit limits can be viewed as financial resources that need to be allocated carefully [30].

However, reinforcement learning may explore risky actions during training. For banking, this can be unsafe. The proposed framework uses rule-based dynamic allocation simulation instead of free reinforcement learning.

Deep learning based financial credit evaluation shows that intelligent models can improve credit assessment [31]. However, deep learning may require larger datasets and stronger explanation tools. The proposed framework uses XGBoost because it provides a practical balance between accuracy and interpretability.

ESG risk management and financial distress research shows that financial risk can be connected with broader sustainability and institutional conditions [32]. This supports future work using macroeconomic and external indicators. Credit risk should not only be based on individual repayment history. It can also be affected by economic conditions, industry risk, and market stress.

Table I compares major research directions related to this study. The table shows that different papers focus on different parts of the problem. Some studies focus on prediction accuracy. Some focus on privacy. Some focus on explainability. Some focus on financial stability. The weakness is that these areas are often separate. A real banking system needs all of them together. This is why the proposed framework combines prediction, explainability, governance, and dynamic action in one system.

Table II explains how the proposed framework is different from earlier approaches. The proposed model is not only a classifier. It includes the full decision process. It starts from borrower data, cleans it, balances it, predicts risk, explains the prediction, checks policy rules, adjusts credit limits, and stores the decision for audit. This makes the framework more complete for practical banking use.

IV. PROBLEM STATEMENT

Modern financial institutions need credit systems that can detect borrower risk and respond quickly. The main problem is that traditional systems are too slow. They depend on fixed review cycles and historical records. In a digital finance environment, this delay can increase loss because risk may grow before the next review date.

Let X_t represent the borrower state at time t . This borrower state may include credit limit, repayment history, bill amount, payment amount, demographic variables, and recent behavior. The model uses this borrower state to estimate the probability of default. If the probability is high, the system must decide whether the credit limit should be reduced or sent for manual review.

The proposed framework must satisfy four conditions. First, borrower data must be cleaned and anonymized before analysis. Second, the model must identify risky borrowers with acceptable accuracy. Third, credit limit decisions must be checked by policy rules before execution. Fourth, every decision must be explainable and stored for audit.

TABLE I
SUMMARY OF RELATED WORK BY METHODOLOGY

Ref.	Full Study Focus	Method or Technology	Experiment Context	Main Contribution	Main Limitation
[13]	Hierarchical Fusion Transformer for Interbank Credit Rating	Transformer-based credit risk modeling	Interbank credit rating and risk relation analysis	Captures complex relationships in financial networks	Complex model and difficult interpretation
[15]	Credit Scoring Prediction Using Deep Learning Models	Deep learning models	Financial sector credit scoring	Learns nonlinear borrower behavior	High computational cost and lower transparency
[17]	Credit Risk Assessment Using Ensemble Models and Explainable AI	Ensemble learning and explainable AI	Structured borrower risk classification	Improves prediction and model interpretation	Does not fully handle dynamic credit action
[19]	Federated Graph Learning-Based Dynamic Risk Modeling	Federated graph learning	Financial institution risk network modeling	Supports privacy-preserving risk analysis	Real-time synchronization can be difficult
[20]	FinTech Innovation and Financial Stability	Financial technology stability analysis	Digital financial systems	Explains the stability impact of FinTech innovation	Does not provide an agentic credit limit model
[25]	Machine and Deep Learning for Credit Scoring with Compliance	Machine learning and compliance design	Credit scoring under regulatory needs	Connects prediction with compliance requirements	Needs stronger operational execution design

TABLE II
COMPARISON OF PROPOSED FRAMEWORK WITH EXISTING APPROACHES

Framework or Approach	Data Processing	Risk Prediction	Explainability	Governance	Privacy Support	Dynamic Limit Allocation	Audit Trail	Real-Time Focus
Deep Learning Credit Scoring [15]	Yes	Yes	Partial	No	No	No	No	Partial
Explainable Ensemble Credit Risk [17]	Yes	Yes	Yes	Partial	No	No	Partial	Partial
Federated Financial Risk Model [19]	Yes	Yes	Partial	Partial	Yes	No	Partial	Partial
Compliant Credit Scoring Model [25]	Yes	Yes	Yes	Yes	Partial	No	Partial	Partial
Multi-Agent Resource Allocation [30]	Partial	Partial	No	Partial	No	Yes	No	Yes
Proposed Framework	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

The main research problem is to design a framework that can predict borrower risk and convert that prediction into a safe dynamic credit limit decision.

$$J = \sum_{i=1}^B (Default_i \cdot L_i) + \alpha(L_{max} - L_{effective}) \quad (1)$$

Equation 1 represents the objective of the proposed framework. The first part, $Default_i \cdot L_i$, represents expected exposure from borrower i . If the borrower has a high default probability and a high credit limit, the expected risk is larger. The summation over B borrowers represents total portfolio exposure. The second part, $\alpha(L_{max} - L_{effective})$, prevents the system from reducing credit too aggressively. The term α controls the balance between bank safety and customer access to credit. This is important because a good system should reduce risk without unfairly blocking safe borrowers.

V. PROPOSED METHODOLOGY

A. Monitoring Agent

The Monitoring Agent is the first active component of the proposed framework. Its main role is to collect borrower information from available data sources. In this study, the UCI Default of Credit Card Clients dataset is used as the main input. In a real banking system, this agent would also receive live transaction streams, repayment updates, and digital banking activity.

This agent is needed because borrower risk changes over time. If the system only checks old data, it may miss new

risk signals. The Monitoring Agent makes the system more active by continuously preparing new borrower states for risk evaluation. It sends the borrower state to the preprocessing layer for cleaning and standardization.

B. Preprocessing Layer

The preprocessing layer cleans the data before it is used by the model. Financial datasets often include noise, inconsistent values, or variables with different scales. If raw data is used without cleaning, the model may produce unreliable results. The preprocessing layer handles this issue by preparing the dataset in a consistent format.

In this study, preprocessing includes checking column names, separating the target variable, removing the ID column, and preparing the feature matrix. The target variable is default payment next month. The feature variables include credit limit, sex, education, marriage, age, repayment status, bill amount, and payment amount.

C. SMOTE Class Balancing

Class imbalance is a common issue in credit risk datasets. Usually, there are more safe borrowers than default borrowers. If the model is trained on this unbalanced data, it may learn to predict most borrowers as safe because that gives high apparent accuracy. This is dangerous because the main purpose of a credit risk system is to identify risky borrowers.

SMOTE is used to solve this issue. It creates synthetic examples of the minority class. It does not simply copy the same risky borrower records. Instead, it creates new examples

using patterns found in existing minority class data. This helps the model learn risky behavior more clearly.

D. Feature Engineering and Recursive Feature Elimination

Feature engineering is used to identify the most useful borrower variables. Not every column has the same importance. Some variables strongly affect credit risk, while others have weaker impact. In this dataset, repayment status variables such as PAY_0 and PAY_2 are very important because they show recent repayment behavior.

Recursive Feature Elimination is used to support feature selection. It removes weaker variables step by step and keeps the stronger predictors. This reduces model complexity and makes the reasoning process easier to explain. It also helps the model focus on variables that matter most for real-time credit decisions.

E. Reasoning Agent

The Reasoning Agent is the main prediction component. It uses XGBoost to calculate borrower default probability. XGBoost was selected because it works well with structured tabular data and can model nonlinear relationships between borrower features.

$$y^* = \arg \max_y P(y|x) \quad (2)$$

Equation 2 shows how the Reasoning Agent selects the predicted borrower class. Here, x represents the borrower feature vector. The value $P(y|x)$ represents the probability that the borrower belongs to class y . The predicted class y^* is the class with the highest probability. In this study, the classes are safe borrower and risky borrower.

The Reasoning Agent does not directly change the credit limit. It only produces the risk prediction and probability. This is important because prediction and action should be separated. The action must still pass through policy validation.

F. Explainability Agent

The Explainability Agent explains the model output. It identifies which borrower variables had the most influence on the prediction. In this study, the feature importance graph is used to understand global model behavior. In a full system, SHAP values could be used to explain each individual borrower decision.

This agent is needed because financial AI systems must be transparent. If a borrower is marked risky, the bank should know whether the decision was mainly caused by repayment delay, high bill amount, low payment amount, or another variable. This makes the system easier to audit and more acceptable for regulated banking use.

G. Action Agent

The Action Agent converts the risk prediction into a credit limit decision. If the predicted risk is below the threshold, the system maintains the credit limit and continues monitoring. If the predicted risk is above the threshold, the system sends the

action to the policy guard for validation. If the policy guard approves the action, the credit limit is reduced.

This component is what makes the framework different from normal credit scoring models. Normal models stop at prediction. The proposed system shows how the prediction can be used to guide financial action.

H. Governance and Policy Guard

The policy guard checks whether the suggested action is safe. For example, it can prevent the system from reducing a credit limit below a minimum allowed level. It can also send very sensitive cases to manual review. This reduces the risk of unsafe automation.

The governance layer also creates a record of the decision. It stores the risk score, key reasons, selected action, and timestamp. This supports audit reporting and compliance review.

VI. SYSTEM ARCHITECTURE AND WORKFLOW

Figure 1 presents the component architecture of the proposed framework. The left side shows the external input layer. This includes the UCI credit dataset, transaction stream, and public FinTech API feeds. These inputs represent different types of borrower and financial signals. The next layer is the Python processing pipeline, which performs data cleaning, SMOTE balancing, and feature selection.

The center of the architecture is the agentic AI core. This part contains the Monitoring Agent, Reasoning Agent, Explainability Agent, and Action Agent. These agents do not work separately. They share information through a common risk state and audit memory. The Monitoring Agent checks data changes. The Reasoning Agent calculates risk. The Explainability Agent produces reasons. The Action Agent prepares credit limit decisions.

The right side of the architecture is the governance and output layer. This layer contains the policy guard, dynamic limit allocation API, compliance dashboard, and human audit report. This design shows that the system does not allow uncontrolled automation. Every important action must pass through governance rules before execution.

Figure 2 shows the workflow of borrower data processing and dynamic credit limit adjustment. The process begins when borrower data is ingested. The data is anonymized and cleaned to protect privacy and remove noise. SMOTE balancing is then applied to improve minority class learning. After that, the system creates the feature vector and sends it to the reasoning layer.

The reasoning layer selects key variables and applies XGBoost inference. The model produces the default probability. If the probability is below the threshold, the borrower limit is maintained and monitoring continues. If the probability is above the threshold, the policy guard validates the action. If the action is safe, the Action Agent adjusts the limit and logs the decision. If the action is not safe, the case is sent for manual review.

Fig. 1. UML Component Architecture — Agentic AI Credit Risk Assessment

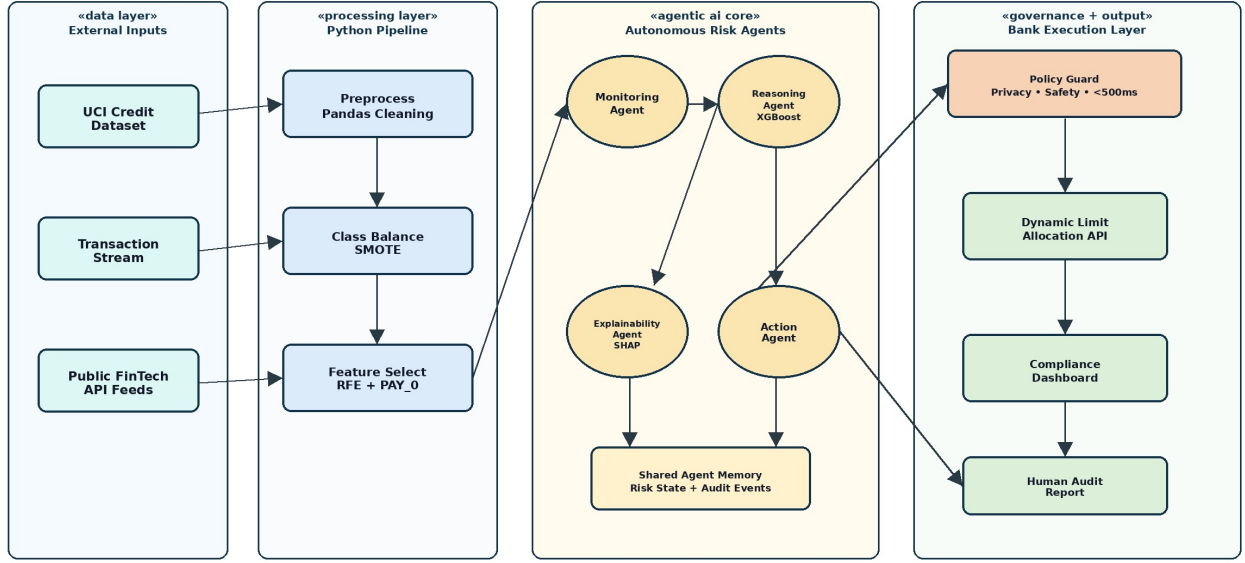


Fig. 1. UML component architecture for Agentic AI credit risk assessment.

VII. IMPLEMENTATION DETAILS

A. Experimental Environment

The framework was implemented using Python 3.12 in a notebook environment. The experiment used common machine learning libraries such as Pandas, NumPy, Scikit-learn, Imbalanced-Learn, XGBoost, Matplotlib, and Seaborn. Pandas and NumPy were used for data handling. Imbalanced-Learn was used for SMOTE. XGBoost was used for classification. Matplotlib and Seaborn were used to generate the graphs.

B. Dataset Description

The UCI Default of Credit Card Clients dataset was used for experimentation. The dataset contains 30,000 borrower records. It includes credit limit, demographic variables, repayment status variables, bill amount variables, payment amount variables, and the default payment target. The target column shows whether the borrower defaulted in the next month.

The dataset is suitable for this study because it contains both historical repayment behavior and borrower financial information. This makes it possible to test whether recent repayment behavior can predict future credit risk.

C. Training Process

The dataset was divided into training and testing sets using an 80:20 ratio after balancing. SMOTE was applied to handle class imbalance. The model was trained using XGBoost with 100 estimators, a learning rate of 0.1, and a maximum depth of 5. These settings were selected to keep the model strong but not too complex.

D. Evaluation Metrics

The model was evaluated using accuracy, precision, recall, F1-score, confusion matrix, feature importance, and ROC-AUC. Accuracy shows the overall correct classification rate. Precision shows how reliable the positive predictions are. Recall shows how many actual cases were correctly detected. F1-score balances precision and recall. ROC-AUC measures the ability of the model to separate safe and risky borrowers across thresholds.

VIII. RESULTS AND DISCUSSION

This section explains the experimental results in detail. The results answer five main questions. First, did SMOTE improve class balance. Second, how well did the model classify safe and risky borrowers. Third, which features had the strongest effect on prediction. Fourth, how strong was the model across different thresholds. Fifth, can the risk prediction be converted into dynamic credit limit action.

Fig. 2. UML Activity Workflow — Real-Time Credit Limit Adjustment

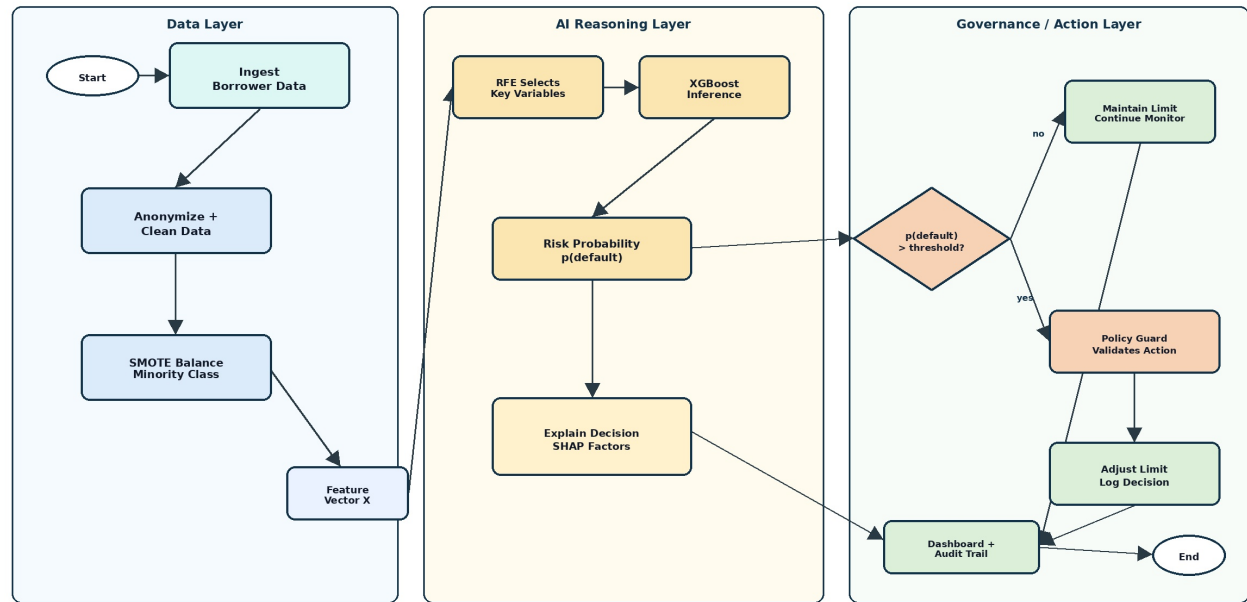


Fig. 2. UML activity workflow for real-time credit limit adjustment.

A. SMOTE Impact Analysis

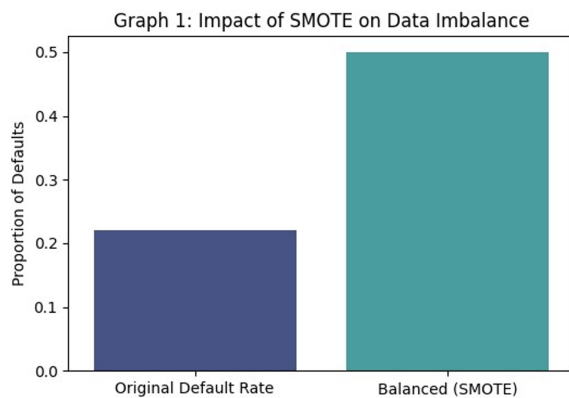


Fig. 3. Impact of SMOTE on class distribution.

Figure 3 shows the impact of SMOTE on the dataset. The first bar shows the original default rate. It is much lower than the balanced SMOTE class distribution. This means the original dataset had fewer default cases than safe cases. This is normal in credit datasets because most borrowers do not default.

The problem is that a model trained on such data may learn to predict the majority class too often. It may classify many borrowers as safe because safe borrowers are more common in the data. This can make the model look accurate, but it may fail

at identifying risky borrowers. In credit risk, this is dangerous because missing risky borrowers can create financial loss.

After SMOTE, the default class becomes better represented. This helps the model learn patterns of default behavior. The purpose of SMOTE is not to change the real world default rate. Its purpose is to improve training quality. The result shows that SMOTE was useful before training the XGBoost Reasoning Agent.

B. Confusion Matrix Evaluation

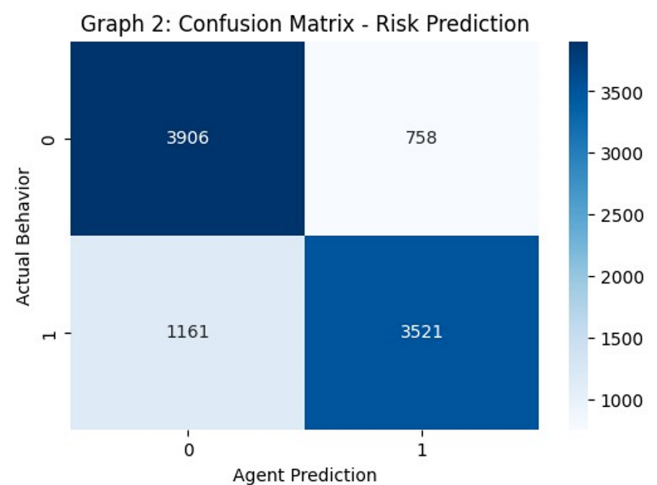


Fig. 4. Confusion matrix for borrower risk prediction.

Figure 4 shows the confusion matrix of the Reasoning Agent. The model correctly identified 3906 safe borrowers and 3521 risky borrowers. These are the correct predictions. The model also produced 758 false positives and 1161 false negatives.

A false positive means the model predicted a borrower as risky when the borrower was actually safe. This can create customer dissatisfaction because a safe borrower may receive a lower credit limit. A false negative means the model predicted a borrower as safe when the borrower was actually risky. In banking, false negatives are more dangerous because risky borrowers may keep access to high credit limits.

The confusion matrix shows that the model performs reasonably well, but it still needs improvement for false negative reduction. This result is useful because it shows where the framework should improve. In future work, the decision threshold can be adjusted to reduce false negatives. Cost-sensitive learning can also be used so that the model treats missed risky borrowers as more serious errors.

C. Feature Importance Analysis

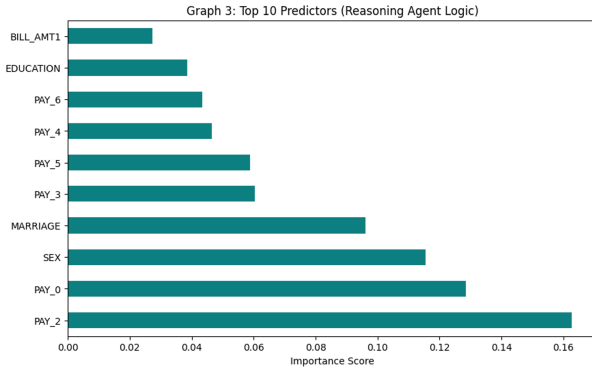


Fig. 5. Top ten predictors identified by the Reasoning Agent.

Figure 5 shows the top ten features used by the XGBoost model. The strongest predictors are repayment status variables, especially PAY_2 and PAY_0. This means recent repayment behavior is more important than many older or static borrower attributes.

This result makes practical sense. A borrower who recently missed payments is more likely to default than a borrower who only has older weak financial signals. Recent repayment status shows current financial stress. This supports the idea that real-time and near real-time monitoring can improve credit risk management.

Other variables such as sex, marriage, education, and bill amount also appear in the feature importance graph. These variables may help the model find additional patterns, but they should be treated carefully. Demographic variables can create fairness concerns if used without control. In a production system, fairness testing would be needed before deployment.

The feature importance result supports the proposed architecture because it proves that the Reasoning Agent is using

meaningful financial signals. It also supports the Explainability Agent because the system can show which variables influenced risk prediction.

D. ROC-AUC Analysis

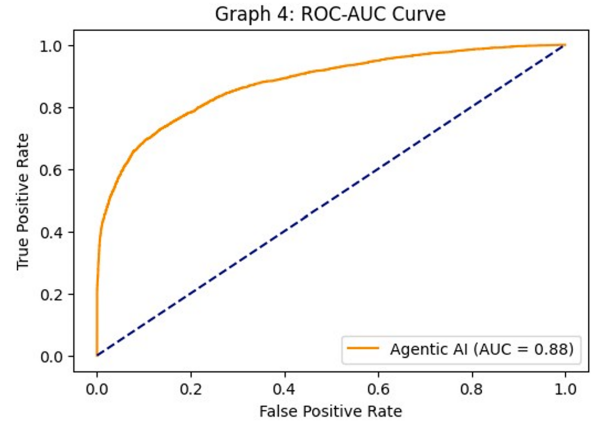


Fig. 6. ROC-AUC curve showing classification strength.

Figure 6 shows the ROC-AUC curve. The ROC curve compares true positive rate and false positive rate across different thresholds. A model that performs randomly would follow the diagonal line. The proposed model is clearly above that line.

The AUC score is 0.88. This means the model has strong ability to separate safe borrowers from risky borrowers. This is important because different banks may use different risk thresholds. A strict bank may lower the threshold to catch more risky borrowers. A more flexible bank may use a higher threshold to reduce false positives. A strong AUC means the model remains useful across different threshold choices.

The ROC-AUC result supports the use of XGBoost as the Reasoning Agent. It shows that the model is not only accurate at one threshold. It has strong ranking ability across many thresholds.

E. Dynamic Credit Limit Allocation

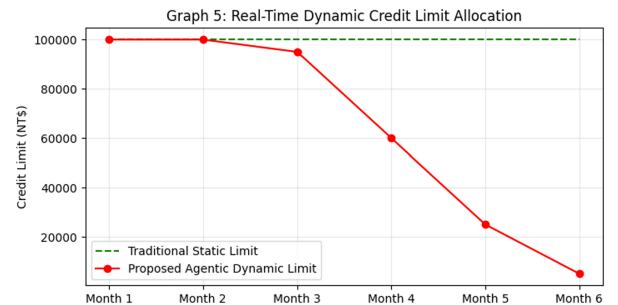


Fig. 7. Simulated dynamic credit limit allocation.

Figure 7 shows the dynamic credit limit simulation. The green dotted line represents the traditional static credit limit.

It stays fixed at NT\$100,000 across all months. This represents a normal banking approach where the limit does not change until a scheduled review.

The red line represents the proposed Agentic AI dynamic limit. It stays stable at first, then decreases when risk increases. The limit moves from NT\$100,000 to NT\$95,000, then NT\$60,000, then NT\$25,000, and finally NT\$5,000. This shows how the system can reduce exposure step by step instead of waiting for a default to happen.

This graph is central to the project. It shows the difference between prediction and action. A normal model may only say that a borrower is risky. The proposed framework uses that risk information to guide credit limit action. This makes the system more useful for real financial operations.

The result is simulation based, not a live banking deployment. This is important to state clearly. The graph demonstrates the logic of the Action Agent. In a real bank, the exact limit reduction would depend on policy, regulation, customer history, and manual review rules.

F. Model Performance Metrics

TABLE III
MODEL PERFORMANCE METRICS

Category	Precision	Recall	F1 Score
Safe Borrowers	0.77	0.84	0.80
Risky Borrowers	0.82	0.75	0.79
Overall Accuracy	0.79		

Table III summarizes the classification performance. The model achieved 79% accuracy. The recall for safe borrowers is 0.84, which means the model correctly identified most safe borrowers. The recall for risky borrowers is 0.75, which means the model correctly identified many risky borrowers, but still missed some.

The risky borrower precision is 0.82. This means that when the model predicted a borrower as risky, the prediction was usually reliable. The F1-scores for both classes are close to each other, which shows balanced performance. This is important because the model should not perform well only on one class.

Overall, the results show that the framework is useful, but it is not perfect. It can support credit risk decisions, but future work should improve risky borrower recall and reduce false negatives.

IX. COMPLEXITY, SECURITY, AND PRIVACY ANALYSIS

The computational complexity of the framework is controlled by using feature selection and XGBoost. Feature selection reduces the number of weak variables and helps the model focus on the strongest predictors. This improves efficiency and reduces unnecessary noise. XGBoost is also suitable because it provides strong results on structured data without needing the large computational cost of deep learning.

Security is important because the Action Agent can affect borrower credit limits. The system must not allow direct

uncontrolled action. This is why the policy guard is included. The policy guard checks whether an action is allowed before it is executed. It can block unsafe decisions, send cases for manual review, and enforce minimum credit limit rules.

Privacy is handled through anonymization before analysis. Borrower identity should not be directly exposed to the model. In a real banking system, tokenization and access control would also be needed. Future versions can use federated learning so that institutions can improve models without sharing raw borrower data.

Auditability is also part of security. Every decision should be logged with the risk score, explanation, action, and timestamp. This makes it easier for auditors to review decisions. It also helps the bank investigate errors if a borrower complains.

The 500 millisecond target is used as a practical response goal. In real-time credit systems, slow action reduces value. If the model detects risk but action is delayed, the system becomes reactive again. The proposed architecture is designed to support fast decision flow, although live deployment would require stronger infrastructure testing.

X. LIMITATIONS

The first limitation is the dataset. The UCI Default of Credit Card Clients dataset is useful for testing, but it is based on older borrower data. It may not fully represent modern digital banking behavior. It does not include mobile wallet usage, gig economy income, real-time transaction streams, or current economic shocks.

The second limitation is that the dynamic credit limit allocation is simulation based. The system shows how risk prediction can be connected to limit action, but it is not deployed inside a real bank. A real deployment would require banking APIs, legal review, customer policy rules, and stronger testing.

The third limitation is the use of demographic variables. Variables such as sex, education, and marriage may improve prediction, but they may also create fairness concerns. A production version should include fairness checks and bias testing before using these variables.

The fourth limitation is false negatives. The model still misses some risky borrowers. This is important because false negatives can create direct financial loss. Future work should improve this through threshold tuning, cost-sensitive learning, and stronger calibration.

XI. FUTURE WORK

While the proposed Agentic AI framework demonstrates strong potential for real-time credit risk assessment and dynamic limit allocation, several directions remain for future research.

First, federated learning should be integrated into the framework. Federated learning would allow multiple financial institutions to train credit risk models together without sharing raw customer data. This would improve privacy and model generalization. It would also help the model learn from a

wider borrower population while protecting sensitive financial records.

Second, graph neural networks should be explored. Borrowers are not always independent. A borrower may be connected to businesses, employers, suppliers, family members, or financial networks. Graph neural networks could help capture these relationships. This would support better understanding of systemic risk and connected borrower behavior.

Third, multi-modal data integration should be added. The current framework uses structured credit card data. Future versions should include real-time transaction streams, macro-economic indicators, employment signals, inflation data, and market stress indicators. This would help the model detect risk in borrowers with limited traditional credit history.

Fourth, formal verification methods should be developed for agentic financial decisions. This means the system should be tested mathematically and operationally to ensure that it does not violate safety rules. Verification would strengthen regulatory compliance and reduce the risk of unsafe automated decisions.

Fifth, the framework should be tested in cross-border lending contexts. Different countries have different regulations, currencies, borrower behaviors, and credit reporting systems. Testing the framework across countries would show whether the architecture can adapt to different financial environments.

Sixth, human-AI collaboration should be studied. Full automation is not always appropriate in banking. Some high-risk cases should be reviewed by human underwriters. Future work should identify when the system should act automatically and when it should ask for human approval.

Finally, longitudinal studies should be conducted. The current study evaluates the model on one dataset. Future research should test the system across full economic cycles, including stable periods, inflation periods, and financial downturns. This would show whether agentic credit systems remain stable over time.

These extensions would move the framework closer to production-ready deployment in regulated financial institutions.

XII. CONCLUSION

This paper proposed an Agentic AI framework for real-time credit risk assessment and dynamic credit limit allocation. The main problem addressed in this study is the delay between credit risk prediction and credit action. Traditional systems often detect borrower risk too late because they depend on fixed review cycles. The proposed framework solves this issue by connecting monitoring, reasoning, explainability, governance, and action.

The framework used SMOTE to handle class imbalance and XGBoost as the main reasoning model. The results showed 79% accuracy and 0.88 ROC-AUC. The confusion matrix showed that the model could identify both safe and risky borrowers with reasonable performance. The feature importance graph showed that recent repayment behavior was the strongest signal for default prediction. The dynamic limit simulation

showed how credit exposure can be reduced when borrower risk increases.

The study shows that credit risk management should not stop at prediction. A useful system must also explain decisions and connect predictions with safe operational action. The proposed framework is not a final production system, but it provides a clear structure for building more responsive and explainable credit risk systems.

Future research should improve the model using live transaction data, federated learning, graph neural networks, formal verification, and human-AI collaboration. These improvements can help move agentic credit risk systems toward safe real-world banking deployment.

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